1. **MERGE SORT VARIATIONS:**
    a. Sorting in descending order.
    b. Dividing the array into 3 sections instead of 2.

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2. Implement the divide and conquer technique to find the **sum of all even numbers** in an array.

    Hint:

    Smallest problem:

        for an array of size 1,

            i) even_sum= 0, if that element is odd.

            ii) even_sum=element, if that element is even.

    Merging strategy:

        even sum of merged_array =even sum of left array + even sum of right array

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3. Implement the divide and conquer technique to find the **second maximum number** in an array.

Hint:

Base case:

    For array of size 2, larger element is the max, smaller one is the second max.

Merging strategy:

    Each smaller problems will return (Max, second_max)

    (Max1, second_max1) = solution of the left subarray.

    (Max2, second_max2) = solution of the right subarray.

    Solution of the merged array should be the maximum and second maximum of Max1, Max2, second_max1, and second_max2

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4. Solve problem 4 but this time, find the **second Minimum number.**

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5. Calculate b^n.

Hint:

b^n = b^(n/2) * b^(n/2) ——————— If n is even.

b^n = b^(n/2) * b^(n/2)*b ——————— If n is odd.

So, calculate b^n/2, then simply multiply it with itself to find b^n.

If n is odd, multiply an extra b.

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6. Implement the divide and conquer technique to find the **multiple of 3 or 5** in an array.

Hint: divide the array to 1 element.

Base case: if the element is a multiple of 3 or 5, return count=1, otherwise return count=0;

Merge: just sum of leftcount and rightcount.

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7. You are given an array of temperatures recorded over a series of days on an island. Some days may have negative temperatures (freezing temperatures), while others may have positive temperatures. Your task is to **find the consecutive days, where the sum of temperatures is the minimum.** The temperatures recorded over the last 10 days are: Temperatures = [3, -2, 5, -4, 6, -1, -2, 3, -5, 2]. Show the detailed simulation to find the result.

Hint:

Minimum sum subarray instead of maximum sum subarray

A trick:

You can simply convert a MIN-algo (ascending sort, max subarray sum) into a MAX-algo (descending sort, min array sum) by doing the following:

1. Interchange MAX and MIN function (if there is a max() in your code, convert it into min() and vice versa)
2. Interchange '>' and '<' for array elements (in your code, if there is a line comparing two array element, just reverse the comparison)
3. Interchange INT_MIN and INT_MAX (if there is a -100000000, make it plus. Also vice versa)